

AI-Augmented Fraud & Phishing Message Detector for Pakistan

(SMS/WhatsApp Security Assistant) — Final Project Report

AI and Automation Internship — THE ARZENS

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1. PROJECT OVERVIEW

This project delivers a working, deployed web application that helps non-technical users in Pakistan assess whether an SMS or WhatsApp message is likely fraudulent. Users paste a suspicious message and receive an instant risk score, a plain-language explanation, and an auto-drafted fraud report they can forward to PTA or the impersonated organization. The tool is designed as risk-screening support, not a definitive verdict, and always encourages independent verification.

The core contribution over existing (largely English-only) phishing detectors is native support for English, Urdu, and Roman Urdu — the mix of languages most commonly used in real scam messages targeting Pakistani mobile users.

2. TECHNICAL APPROACH

The system combines two independent detection layers so that either one catching a scam is enough to raise the alarm, rather than averaging a confident signal down:

2.1 RULE-BASED SECURITY CHECKS

- Typosquatting detection — edit-distance comparison of message links against a seed list of ~40 legitimate Pakistani bank, telecom, courier, and e-commerce domains
- WHOIS domain-age lookup — flags domains registered within the last 90 days
- SSL certificate validation — flags missing or invalid HTTPS certificates
- URL structure analysis — flags IP-based links, known shortener services, excessive subdomains, and '@' redirect tricks

2.2 LLM-BASED LANGUAGE ANALYSIS

A large language model (accessed via the Groq API) reads the message text directly and evaluates it for social-engineering patterns: urgency and threats, impersonation of a known organization, requests for OTP/PIN/CNIC, prize/lottery framing, and pressure to click a link. The prompt was tuned with few-shot examples across all three supported languages, and explicitly instructed not to over-flag urgency alone — a real flight-delay notice or a real OTP message should not be treated the same as a phishing attempt just because both are time-sensitive.

2.3 COMBINED RISK SCORING

The overall score takes the stronger of the rule-based and LLM signals as its base, with a smaller bonus when both agree. This ensures that a scam caught only by rule-based checks (e.g., a typosquatted domain with no urgent language) or only by the LLM (e.g., a scam using a legitimate-looking shortener with no rule-based red flags) is still correctly flagged as high risk, rather than being diluted by averaging with a quiet second signal.

3. TECHNOLOGY STACK

- Frontend/App: Python, Streamlit
- LLM: Groq API (openai/gpt-oss-120b), accessed via an OpenAI-compatible client
- Security checks: python-whois, Python ssl/socket, difflib/edit-distance, tldextract

- Development: Google Colab (prototyping and iterative testing)
- Deployment: GitHub + Streamlit Community Cloud (free tier)

4. TESTING & RESULTS

The system was evaluated in two ways: (1) a structured 20-message labeled test set spanning English, Roman Urdu, and Urdu script — including deliberately ambiguous and legitimate-but-urgent messages to check for false positives — run through an automated evaluation script; and (2) live manual testing on the deployed application, shown below.

4.1 AMBIGUOUS CASE — CORRECT CALIBRATION

A real-style bank debit alert asking the recipient to "call for queries" scored 30/100 (Likely Safe) rather than being over-flagged, while the explanation still noted appropriate caution about the phone-based request. This demonstrates the system distinguishes between routine notifications and genuine social-engineering requests, rather than reacting to urgency or money-related keywords alone.



Figure 1: A real-style debit alert correctly scored as low risk, with appropriate caution noted in the explanation.

4.2 CROSS-LANGUAGE PHISHING DETECTION

The identical UBL account-suspension phishing scenario was tested in English, Roman Urdu, and Urdu script. All three were correctly identified as high risk (92-96/100), confirming the multilingual design goal from the original proposal.



Fig 2: English — 96/100



Fig 3: Roman Urdu — 96/100



Fig 4: Urdu script — 92/100

4.3 AUTO-DRAFTED FRAUD REPORT

For any flagged message, the system generates a ready-to-send report including the risk assessment, detected language, original message content, claimed organization, and a downloadable .txt file — reducing friction for users who might otherwise not bother reporting a scam.



Figure 5: Auto-generated fraud report draft, ready to forward to PTA or the impersonated organization.

5. CHALLENGES ENCOUNTERED

- WHOIS data is incomplete or unavailable for some domains — handled with graceful fallback so the app never crashes, only shows 'unavailable' for that specific check.
- LLM output is not 100% reliable — mitigated with few-shot prompting and framing all results as risk-screening rather than a definitive verdict.
- Limited local-language training/testing data — addressed by hand-building a labeled multilingual test set and testing manually across scam types; a larger set of real (not synthetic) scam messages remains a priority for future work.
- Deployment platform friction — Hugging Face Spaces changed its free-tier SDK options mid-project (Streamlit removed as a direct SDK, Docker moved behind a paywall, and a Spaces quota limit was hit). The project was successfully redeployed on Streamlit Community Cloud instead, which is purpose-built for Streamlit apps and avoided these constraints.

6. FUTURE IMPROVEMENTS

- Replace the synthetic test set with a larger set of real, self-collected scam messages
- Direct integration with PTA's official complaint submission system
- Browser extension or mobile app for real-time link scanning
- Crowdsourced scam database built from user submissions over time
- Expanded brand-domain list and support for additional regional languages

7. CONCLUSION

The project successfully delivers on its core proposal: a working, deployed, hybrid AI + rule-based fraud detector that handles the multilingual reality of SMS/WhatsApp scams in Pakistan. Testing across languages and scam types shows the combined-signal scoring approach correctly identifies both technically obvious scams (typosquatted domains, suspicious shorteners) and scams that rely purely on social-engineering language, while avoiding false positives on legitimate urgent messages. The live application, source code, and testing scripts are available in the project repository.